Slip21

1. Write an Android Program to Demonstrate Date Picker Dialog in Android.

Project Structure

```
app/
├── java/com/example/datepicker/
│   └── MainActivity.java
└── res/
    └── layout/
        └── activity_main.xml
```

a. XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

    xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:orientation="vertical"

    android:gravity="center"

    android:padding="20dp">

    <TextView

        android:id="@+id/textViewDate"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Selected Date"

        android:textSize="20sp"
```

```
        android:padding="10dp"/>

    <Button
        android:id="@+id/buttonPickDate"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Date"/>
</LinearLayout>
```

b. MainActivity.java

package com.example.datepicker;

```
import androidx.appcompat.app.AppCompatActivity;
import android.app.DatePickerDialog;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import java.util.Calendar;
public class MainActivity extends AppCompatActivity {
  TextView textViewDate;
  Button buttonPickDate;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
```

```
    setContentView(R.layout.activity_main);

    // Link UI components

    textViewDate = findViewById(R.id.textViewDate);

    buttonPickDate = findViewById(R.id.buttonPickDate);

    // Button click event

    buttonPickDate.setOnClickListener(new View.OnClickListener() {

      @Override

      public void onClick(View v) {

        // Get current date

        Calendar calendar = Calendar.getInstance();

        int year = calendar.get(Calendar.YEAR);

        int month = calendar.get(Calendar.MONTH);

        int day = calendar.get(Calendar.DAY_OF_MONTH);

        // Create DatePickerDialog

        DatePickerDialog datePickerDialog = new DatePickerDialog(

            MainActivity.this,

            new DatePickerDialog.OnDateSetListener() {

              @Override

              public void onDateSet(android.widget.DatePicker view, int year, int month, int
dayOfMonth) {


                // Set selected date to TextView

                textViewDate.setText(dayOfMonth + "/" + (month + 1) + "/" + year);

              }

            },
```

```
                year, month, day);

            // Show dialog

            datePickerDialog.show();

        }

    });

  }

}
```

2. Create table Game(no,name,type, no_of_players). Create Application to perform the following operations.

i) Update no_of_players to four where game is Badminton.

ii) Display all the records.

Project Structure

```
app/
 ├── java/com/example/gameapp/
 │    ├── MainActivity.java
 │    ├── DBHelper.java
 ├── res/layout/
 │    └── activity_main.xml
 └── AndroidManifest.xml
```

1. Database Helper Class

DBHelper.java

```
package com.example.gameapp;

import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
```

```
public class DBHelper extends SQLiteOpenHelper {

    public DBHelper(Context context) {
        super(context, "GameDB", null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        // Create table
        db.execSQL("CREATE TABLE Game(no INTEGER PRIMARY KEY, name TEXT, type TEXT,
no_of_players INTEGER)");

        // Insert sample data
        db.execSQL("INSERT INTO Game VALUES(1,'Cricket','Outdoor',11)");
        db.execSQL("INSERT INTO Game VALUES(2,'Football','Outdoor',11)");
        db.execSQL("INSERT INTO Game VALUES(3,'Badminton','Indoor',2)");
        db.execSQL("INSERT INTO Game VALUES(4,'Chess','Indoor',2)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS Game");
        onCreate(db);
    }

    // Update Badminton players to 4
    public void updateGame() {
        SQLiteDatabase db = this.getWritableDatabase();
        db.execSQL("UPDATE Game SET no_of_players=4 WHERE name='Badminton'");
    }

    // Fetch all records
    public Cursor getData() {
        SQLiteDatabase db = this.getReadableDatabase();
        return db.rawQuery("SELECT * FROM Game", null);
    }
}
```

Explanation (DBHelper.java)

- SQLiteOpenHelper → Helps manage database
- onCreate() → Creates table + inserts data
- execSQL() → Executes SQL queries
- updateGame() → Updates Badminton players to 4
- getData() → Returns all records using Cursor

2. activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:orientation="vertical"
  android:padding="16dp"
  android:layout_width="match_parent"
  android:layout_height="match_parent">
  <Button
    android:id="@+id/btnUpdate"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Update Badminton Players"/>
  <Button
    android:id="@+id/btnDisplay"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
```

```
        android:text="Display Records"/>
    <TextView
        android:id="@+id/txtResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="16sp"
        android:paddingTop="20dp"/>
</LinearLayout>
```

3. MainActivity.java

```
package com.example.gameapp;


import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    DBHelper db;
    Button btnUpdate, btnDisplay;
    TextView txtResult;

    @Override
```

```
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    // Initialize components
    db = new DBHelper(this);
    btnUpdate = findViewById(R.id.btnUpdate);
    btnDisplay = findViewById(R.id.btnDisplay);
    txtResult = findViewById(R.id.txtResult);
    // Update Button Click
    btnUpdate.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            db.updateGame();
            txtResult.setText("Badminton players updated to 4");
        }
    });
    // Display Button Click
    btnDisplay.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            Cursor c = db.getData();
            String data = "";
            while (c.moveToNext()) {
                data += "No: " + c.getInt(0) + "\n";
```

```
            data += "Name: " + c.getString(1) + "\n";
            data += "Type: " + c.getString(2) + "\n";
            data += "Players: " + c.getInt(3) + "\n\n";
        }
        txtResult.setText(data);
      }
    });
  }
}
```